



Environmental Product Declaration

Schindler 5000 and 5000 Plus

Program:	The International EPD® System EPD International AB www.environdec.com
EPD registration number:	EPD-IES-0016448
Published:	2025-05-22
Valid until:	2030-05-22
Product group classification:	UN CPC 4354



in accordance to ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021

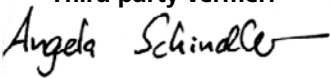
An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com.



Program-related information and verification

Reference year for data:	2019–2023
Geographical scope:	China
Product category rules (PCR):	EN 15804:2012+A2:2019/AC:2021 as Core PCR PCR 2019:14 Construction Products, version 1.3.4 C-PCR-008 Lifts (to PCR 2019:14), Version 1.0.0, 2025-04-08
PCR review was conducted by:	The Technical Committee of the International EPD® System. See www.environdec.com/about-us/the-international-epd-system-about-the-system for a list of members. Review chair: Gorka Benito Alonso. The review panel may be contacted via the Secretariat www.environdec.com/contact-us .
EPD owner:	Schindler Management Ltd Zugerstrasse 13 6030 Ebikon Switzerland The EPD Owner has sole ownership, liability and responsibility for the data contained within this EPD
LCA author:	Carbotech AG St. Alban-Vorstadt 19 4052 Basel Switzerland www.carbotech.ch
Program operator:	EPD International AB info@environdec.com
Procedure for follow-up during EPD validity involves third party verifier:	No

Verification:

CEN standard EN15804 serves as the core PCR
Independent verification of the declaration and data, according to EN ISO 14025:2010 ☐ Internal ☒ external
Third party verifier:  Angela Schindler, Umweltberatung und Ingenieurdienstleistungen Approved by The International EPD(R) system

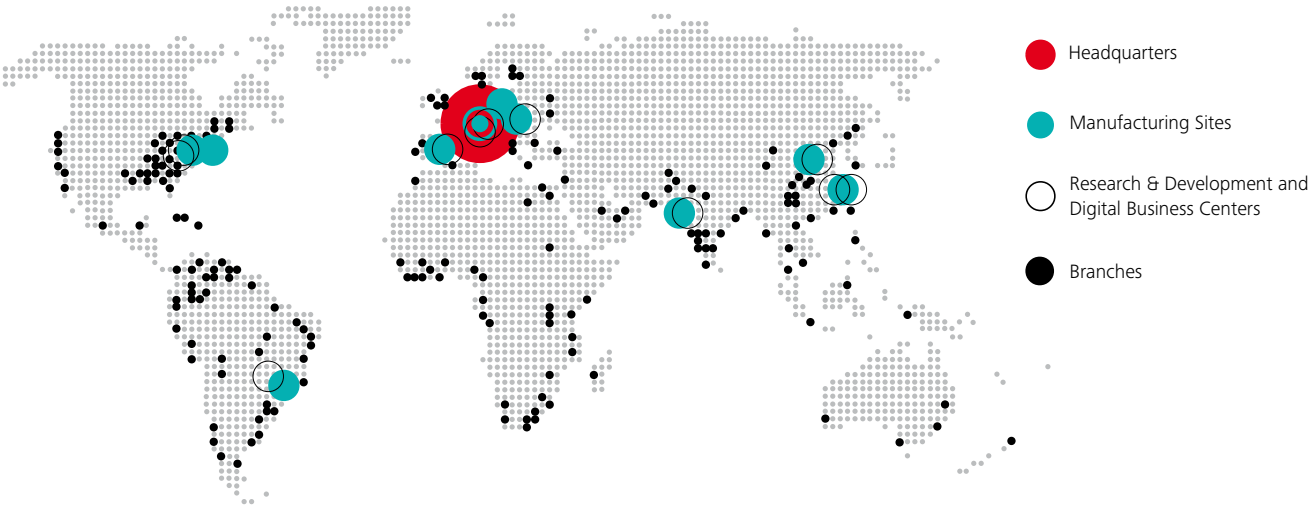
About Schindler

Founded in Switzerland in 1874, the Schindler Group is a leading global provider of elevators, escalators, and related services. Schindler mobility solutions move more than one billion people every day all over the world.

Behind the company’s success are over 70,000 employees in more than 1,000 branches in over 100 countries throughout Europe, North & South America, Asia-Pacific, and Africa with manufacturing plants strategically located in Europe, Brazil, USA, China, and India.

Schindler manufactures, installs, services, and modernizes elevators, escalators, and moving walks for almost every type of buildings worldwide. Schindler’s offerings range from cost-effective solutions for low-rise residential buildings to sophisticated access and transport management concepts for skyscrapers.

Schindler moves people and materials, and connects vertical and horizontal transport systems through intelligent mobility solutions driven by green and user-friendly technologies. Schindler products can be found in many well-known buildings across the globe, including residential and office buildings, airports, shopping centers / retail establishments, and buildings with special requirements.



A network of more than 1,000 branches in over 100 countries.

Comparability between EPDs based on this c-PCR-008 (to PCR 2019:14) and EPDs based on PCR 2015:05 is not conceivable and shall be avoided. Any comparability of this kind shall be considered as false and misleading the EPD user. EPDs within the same product category but registered in different EPD programmes may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison.

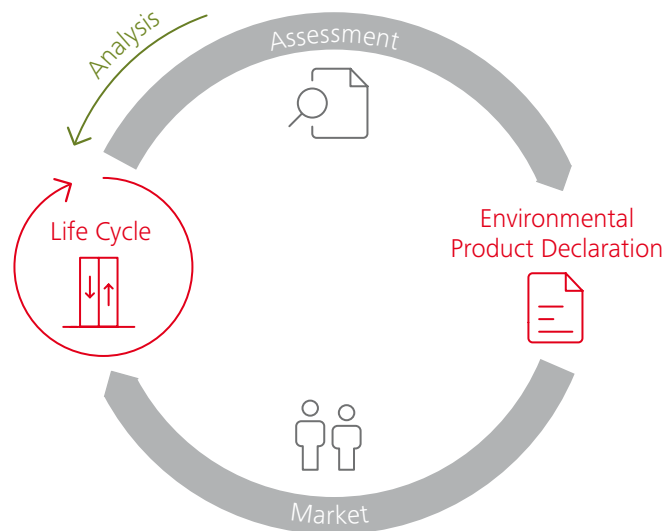
We Elevate... Sustainability

Schindler’s commitment to sustainability is enshrined in our Corporate Sustainability Policy, which defines our approach to sustainability based on four pillars – People, Product, Planet, and Performance – and the journey we have embarked on regarding key sustainability challenges.

Sustainability is a dual commitment for Schindler: we want to fulfill our vision of leadership in urban mobility solutions and strive to optimize our environmental impact while investing in people and society. Schindler has demonstrated this commitment by achieving the ISO 9001/14001 certification in 2020.

Mobility is essential in the world in which we live and work. Every day, more than two billion people all over the world place their trust in Schindler. That is why we are committed to continuously improving the environmental impact of our products and services along the whole life cycle.

Since 1874, Schindler has grown around the world and is recognized as a responsible corporate citizen. We firmly intend to continue evolving along this path with a global perspective on sustainability and a focus on the most relevant key performance indicators.



From design to recycling

From the first sketches in design, right through to disposal and recycling, environmental assessment considerations are an integral part of the Schindler product development process. This process rigidly follows the ISO 14040 standard and is embedded in the ISO 14001 Environmental Management System, which is applied by Corporate Research & Development, along with many other internal functions, to provide transparency in all phases.

Life Cycle Assessment (LCA)

Schindler conducts Life Cycle Assessments of its products, in order to continuously improve their environmental performance. A holistic approach is applied all the way from initial product development through to the continuous product improvement initiatives.

Environmental Product Declaration (EPD)

The EPD provides verified information on the environmental impact of a product. The declaration is based on a comprehensive LCA and follows the ISO 14025 guideline. A complex issue is made understandable.

Product Category Rules (PCR)

Product Category Rules define the rules and requirements for EPDs of a certain product category. They are a key part of ISO 14025 as they enable transparency and comparability between EPDs.

Thinking globally. Acting locally.

Local production

With manufacturing plants strategically located in Europe, Brazil, USA, China, and India, Schindler focuses on local production for the local market. This reduces the environmental impact from shipping and transport around the world.

In China, Schindler has a manufacturing plant in Jiading, outside of Shanghai. With 95% of the components in the Schindler 5000 produced or assembled in China, we strive to use effective and efficient transport methods to get the material to each jobsite to minimize our carbon footprint.

Modular products

Our modular approach to system development enables us to share components across products, including the Schindler 1000, Schindler 3000, Schindler 5000 and Schindler 6000. This enables better sourcing management with our suppliers and sub-suppliers and consolidation of shipments to reduce the environmental impact caused by the transport of material to Schindler manufacturing plants.

By optimizing our logistic activities and manufacturing supplier base, the supply chain in China has substantially reduced the logistic carbon dioxide footprint for the Schindler 5000.

Recyclable packaging

Packaging of the Schindler 5000 is mainly comprised of recyclable material, such as cardboard, paper, PE plastic and wood. This enhanced packaging features a robust and damage-resistant shell to protect our products in transit and on the construction site, while also reducing waste. It has been qualified in a test lab to ensure durability.

The packaging concept has been defined in combination with the installation process and has been designed to support the sequence of activities during the elevator installation. This ensures material remains un-damaged since it can remain packaged and protected until it is required for installation.

Digital processes

To improve our installation process and drive sustainability in the field, Schindler has digitized the installation and commissioning manuals for our fitters. By making these documents available on mobile devices, we have reduced our impact on natural resources, saving 250 metric tons of paper annually.





Ascend to excellence Schindler 5000 and 5000 Plus

Striking, flexible, effective
Offering exceptional adaptability and striking interior options, the Schindler 5000 ensures a custom fit for any setting. Enjoy the perfect blend of form, function, and flexibility with a single solution that suits both commercial and high-end residential projects.

The Schindler 5000 is not just about aesthetics; it's engineered for peak performance. With cutting-edge technology, premium materials, and energy-saving innovations, it sets a new standard for elevator efficiency and passenger comfort.

The Schindler 5000 delivers a swift, smooth, and energy-efficient journey – combining the latest technology, enhanced efficiency and flexible design, for perfect integration into your building.

The Schindler 5000 Plus is a replacement elevator for existing buildings.

Key figures	
Capacity	630 to 1350 kg
Travel height	Up to 80m
Drive	Schindler Traction Media Technology; Synchronous machine with regenerative drive
Speed	1.0 to 2.5 m/s
Number of stops	Up to 40
Car groups	Up to 8 cars
Entrances	One- or two-sided

Perfectly suited to the environment



- Overall system**
- Compact, lightweight, and durable design that optimizes material usage
 - Remote connectivity improves service efficiency and reduces unnecessary trips to the equipment for maintenance and service
 - Available in a space-saving machine-room-less (MRL) design

- Drive**
- Gearless machine for smooth ride quality without requiring oil for lubrication
 - Regenerative frequency converter returns energy to the grid for future use in the building or elevator operation
 - Stable start without high peak current, quickly reaching a low energy consumption level

- Car**
- Ceiling lights, car indicator and landing indicators feature energy-saving LED lights
 - Door drive with stand-by mode for safety and energy conservation
 - Light-weight interior materials improve operational efficiency and energy usage

- Control**
- System switches car lights and ventilation into stand-by mode when not in use
 - Smart operation, down collective and selective collective controls for efficient passenger transportation
 - Halogen-free cabling and wiring, available as an option

- Hoistway**
- Lighter Schindler Traction Media require less energy to operate than traditional steel ropes
 - Updated elevator positioning system eliminates unnecessary trips to reset the system
 - Compared to steel ropes, the STM is oil-free and does not require regular lubrication

Representative unit
based on an average mid-rise residential building in China

Rated load	1000 kg
Speed	1.75 m/s
Travel height	42 m
Number of floors / entrances	15 / 1

In case of major deviations to the given configuration, please contact Schindler to anticipate the impact.

Car W/D/H	1600x1500x2500 mm
Door W/H	900x2100 mm
Operation days per year	365
Usage category	3
Reference service life (RSL)	25 years

Elevator life cycle insights

System boundary

This EPD covers the full life cycle with a cradle to grave approach. The PCR focuses on four main stages. The Product stage (A1-A3) includes the raw material extraction and production, transport to the manufacturing site (primarily by truck), and manufacturing and assembly of components, considering the demand of energy, auxiliary and operational materials, and packaging. The Construction process stage (A4-A5) includes the transportation to the installation site by truck and the installation, considering the energy demand and auxiliary materials including related Volatile Organic Compound (VOC) emissions. The Use stage (B1-B7) includes the maintenance, considering the transportation of employees to the installation site and auxiliary materials, including related

VOC emissions and replacement parts production and energy use during operation and standby during the reference service life. All other modules are not relevant and modernization is not foreseen. The End-of-life stage (C1-C4) includes the deconstruction, considering the energy demand and auxiliary materials, the transportation by truck to waste processing facilities, the waste processing, considering sorting, and the waste disposal, considering a scenario with recycling, incineration, and landfill. Finally, the benefits and loads beyond the system boundaries stage (D) includes the potential for recycling by substitution of primary material and energy recovery. Infrastructure and capital goods are included in most background data from ecoinvent. No additional infrastructure and capital goods were considered for own factory balances.

Cut-off criteria

General quality and cut-off criteria were considered, as defined for the evaluation in the PCR and EN 15804. The total mass of the elevator materials considered equals the total mass of the elevator. All inflows and

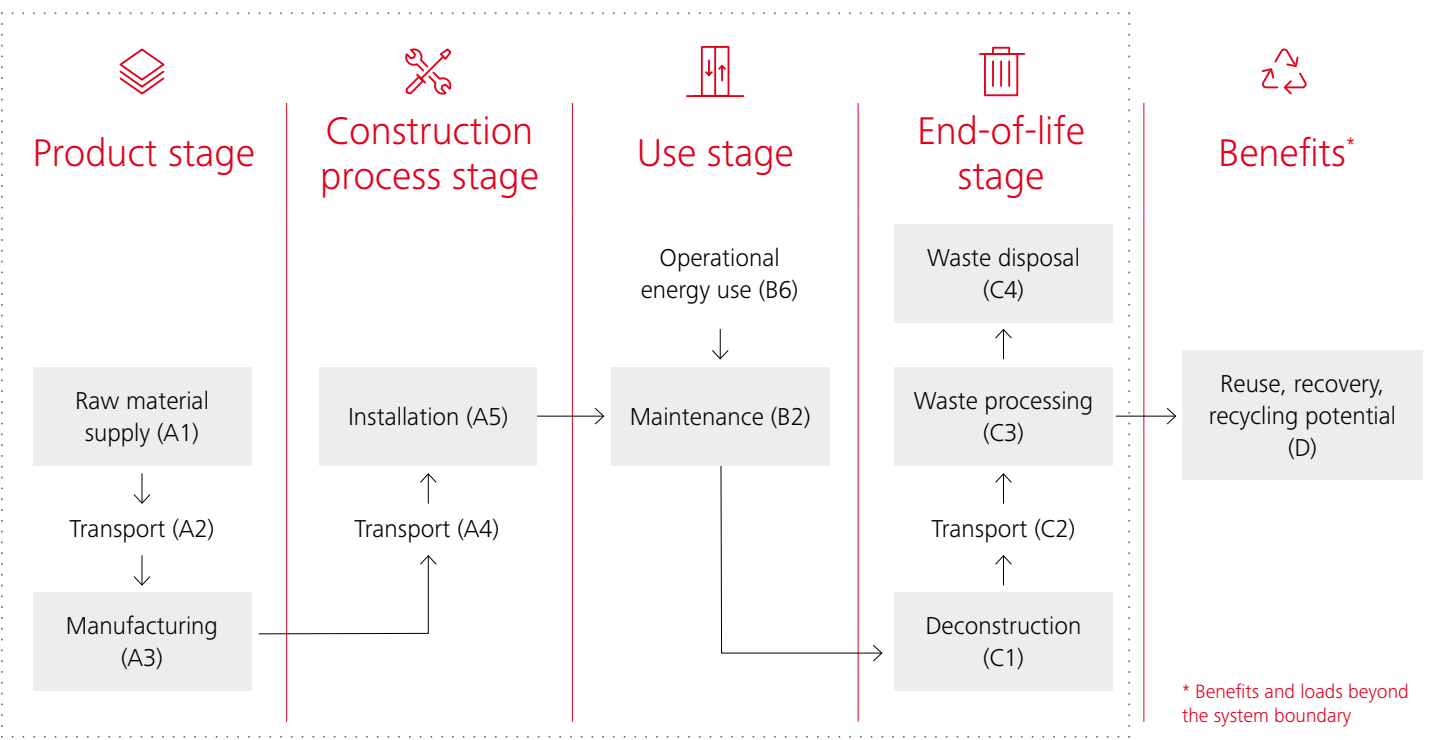
outflows, for which data are mandatory, are included in the calculations. Special emphasis was given to material and energy flows that are known to have a large impact.

		Modules declared	Geography	Share of specific data
Product stage	Raw material supply	A1 ✓	CN/Global	> 90%
	Transport	A2 ✓	Global	
	Manufacturing	A3 ✓	CN/EU*	
Construction process stage	Transport	A4 ✓	CN	> 90%
	Installation	A5 ✓	CN	
Use stage	Use	B1 ND	ND	ND
	Maintenance	B2 ✓	CN/Global	
	Repair	B3 ND	ND	ND
	Replacement	B4 ND	ND	ND
	Refurbishment	B5 ND	ND	ND
	Operational energy use	B6 ✓	CN	
	Operational water use	B7 ND	ND	ND
End-of-life stage	Deconstruction	C1 ✓	CN	
	Transport	C2 ✓	CN	
	Waste processing	C3 ✓	CN/Global	
	Waste disposal	C4 ✓	CN/Global	
Beyond the system boundaries	Reuse, recovery, recycling, potential	D ✓	CN/Global	

This declaration covers “cradle to grave”. All mandatory modules covered in the EPD are marked with a ✓. For non-relevant fields, ND is marked in the table.

*Manufacturing locations in FL

System boundary

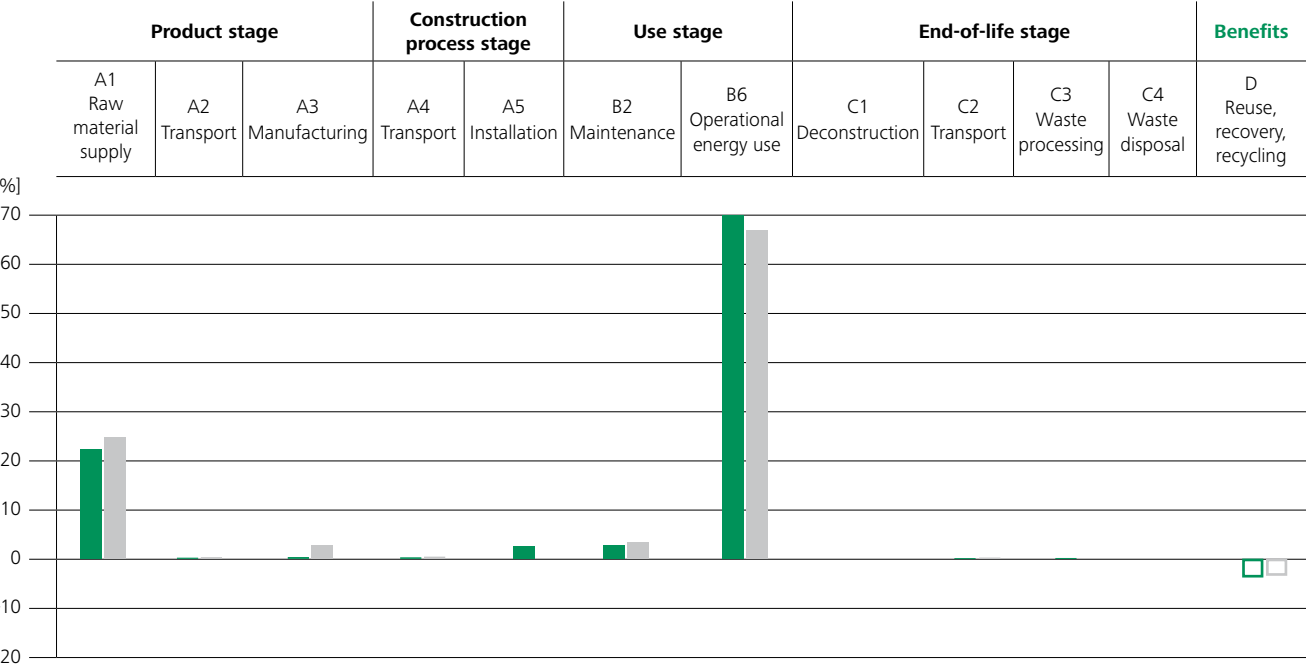


Our mission: reduce emissions

Consolidated impact

Values shown refer to the representative unit of Schindler 5000, as shown on page 7 based on a RSL of 25 years. The most relevant processes, energy and material flows are indicated.

- Climate change total (GWP_{tot})
- Resource use - fossil fuels (ADPF)



Data reflects UC3 results

Environmental impact

In the LCA, impact assessment methods and characterization factors were used at the midpoint level, as requested in the PCR (i.e. without normalization and weighting). Selected core environmental impact categories for this study were global warming (IPCC 2021 100 year horizon), effects on the stratospheric ozone layer (WMO, 2014), acidification (Seppälä et al., 2006), eutrophication (Struijs et. Al 2009b), photochemical ozone creation (Van Zelm et al.), abiotic depletion of elements (CML 2001, baseline, August 2016 version), abiotic depletion of fossil fuels (Guinée et al.), and water deprivation potential (Boulay et al., 2018).

Impacts per functional unit

The PCR defines the following functional unit for product comparison.

The primary purpose of an elevator is to vertically transport goods and passengers. Therefore, for the purpose of this EPD, the functional unit is the result of a load transported over a distance, expressed in tonne - kilometer [tkm].

The Transportation Performance (TP) indicates the total amount of tkm performed by the elevator over the defined service life with an average load, according to ISO 25745-2. For the defined representative unit and a RSL of 25 years, the TP per applied usage category is:

Usage category	Transportation Performance (TP)
3	2535.2 tkm

Summary

Energy rating efficiency has been further improved compared to the previous product generation. In the operations stage, we have achieved a Class A energy efficiency rating for the defined representative elevator. The material supply for the production followed by the energy consumption of the elevator during operation have the biggest environmental impact. The profile of

the impacts of the energy consumption depends on the chosen electricity supply. The Chinese supply mix was considered for the installation in Zhejiang. Further relevant factors are the lifetime of elevator parts and the usage category. With shorter lifetime of parts and lower usage, the portion of materials becomes more important.



*The classification refers to a specific customer configuration as defined for the Life Cycle Assessment, see page 7. Usage pattern, load capacity, customer specific options and site conditions influence the final rating.

Minimizing material, maximizing space

Material that matters

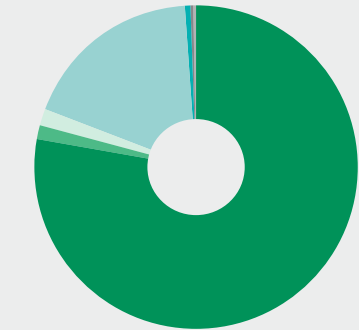
The table and graph below show the resulting material composition of the installed elevator with a total weight of 5994.03 kg, without packaging. It is mainly composed of ferrous metals. The biogenic carbon content in the product is below 5%.

At the end of use almost all material is suitable for recycling. An average material loss of 5% in production was assumed additionally for the consumption of raw material. The Schindler 5000 elevators emit no VOCs or other harmful substances once installed. The cabling and wiring in a Schindler elevator can also be ordered

halogen-free. Hazardous substances are avoided as much as possible, in accordance with REACH, its candidate list and other regulations. However, in some articles used in our products, substances of the candidate list, in particular lead (CAS-No. 7439-92-1), may still exist and are reported in ECHA's SCIP database at <https://echa.europa.eu/de/factsheet/-/factsheet/174759520>.

Used material – an overview

Product components	Weight (kg)	Weight (%)	Post-consumer material weight (%)
● Ferrous metal	4666.87	77.86	unknown
● Non-ferrous metals	84.30	1.41	unknown
● Plastics and rubbers	93.65	1.56	0
● Inorganic materials	1092.75	18.23	0
● Organic materials	24.93	0.42	0
● Lubricants	0.04	0.00	0
● Electric and electronic equipment	24.78	0.41	unknown
● Batteries and accumulators	6.71	0.11	unknown
● Other materials	0.00	0.00	0
Total	5994.03	100	



Packaging material

The table shows the typical composition of material used for packaging in relation to the total weight of the elevator system – once the elevator arrives on the construction site.

Schindler seeks to maximize the transport capacity per pallet for each delivery. Furthermore, almost all materials are suitable for recycling, e.g. paperboard and wood.

Composition of packaging material				
Product components	Weight (kg)	Weight (%)	Weight (%) packaging vs product	Biogenic carbon content (kg C)
Wood*	790.00	90.80	13.18	3.95E+02
Cardboard*	18.00	2.07	0.30	8.28E+00
Plastic	24.00	2.76	0.40	0.00E+00
Steel	38.00	4.37	0.63	0.00E+00
Total	870.00	100	14.51	4.03E+02

*Renewable material

Potential environmental impact

Table of results – core environmental impact UC 3 per tkm															
	EN15804	Product stage				Construction process stage		Use stage		End-of-life stage					Net Benefits
Impact category	Unit	A1	A2	A3	Sum A1–A3	A4	A5	B2	B6	C1	C2	C3	C4	Total	D
GWP _{tot}	kg CO ₂ eq.	5.13E+00	8.42E-02	1.10E-01	5.32E+00	1.02E-01	6.26E-01	6.46E-01	1.60E+01	8.05E-03	6.18E-02	7.21E-02	2.03E-02	2.29E+01	-8.10E-01
GWP _{fos}	kg CO ₂ eq.	5.12E+00	8.41E-02	6.91E-01	5.89E+00	1.02E-01	4.24E-02	6.43E-01	1.60E+01	8.04E-03	6.17E-02	7.20E-02	1.92E-02	2.28E+01	-8.11E-01
GWP _{bio}	kg CO ₂ eq.	5.23E-03	2.05E-05	-5.83E-01	-5.77E-01	2.69E-05	5.83E-01	1.60E-03	2.63E-03	1.32E-06	1.92E-05	4.79E-05	1.16E-03	1.14E-02	8.51E-04
GWP _{luluc}	kg CO ₂ eq.	7.96E-03	3.89E-05	1.51E-03	9.51E-03	5.33E-05	4.26E-06	1.10E-03	6.53E-03	3.28E-06	3.96E-05	1.43E-05	3.24E-06	1.73E-02	8.57E-05
ODP	kg CFC 11 eq.	1.14E-07	1.31E-09	3.59E-09	1.19E-07	1.62E-09	1.30E-10	2.36E-08	3.43E-08	1.72E-11	9.36E-10	1.53E-09	9.65E-11	1.81E-07	-2.13E-08
AP	mol H ⁺ eq.	4.31E-02	3.63E-04	3.60E-03	4.70E-02	4.49E-04	1.06E-04	5.03E-03	8.82E-02	4.44E-05	2.59E-04	7.00E-05	2.92E-05	1.41E-01	-6.02E-03
EP _{fw}	kg P eq.	3.75E-03	6.14E-06	1.52E-04	3.91E-03	8.31E-06	3.26E-06	5.43E-04	3.20E-03	1.61E-06	5.90E-06	7.15E-06	6.65E-07	7.68E-03	-4.92E-04
EP _{mar}	kg N eq.	6.29E-03	1.34E-04	8.50E-04	7.27E-03	1.65E-04	3.93E-05	1.06E-03	1.81E-02	9.12E-06	8.89E-05	2.44E-05	2.81E-04	2.71E-02	-6.52E-04
EP _{ter}	mol N eq.	8.37E-02	1.43E-03	8.48E-03	9.36E-02	1.76E-03	3.88E-04	9.58E-03	1.94E-01	9.73E-05	9.49E-04	2.09E-04	1.17E-04	3.00E-01	-9.53E-03
POCP	kg NMVOC eq.	2.69E-02	4.86E-04	2.48E-03	2.98E-02	6.05E-04	1.27E-04	3.01E-03	5.13E-02	2.58E-05	3.30E-04	6.21E-05	3.91E-05	8.53E-02	-4.82E-03
ADPE*	kg Sb eq.	2.63E-04	2.39E-07	2.60E-06	2.65E-04	3.29E-07	4.06E-08	1.23E-04	6.54E-05	3.29E-08	2.62E-07	4.13E-08	9.16E-09	4.55E-04	-5.70E-05
ADPF*	MJ	5.79E+01	1.17E+00	6.95E+00	6.60E+01	1.45E+00	1.53E-01	8.36E+00	1.56E+02	7.84E-02	8.49E-01	1.65E-01	8.37E-02	2.33E+02	-7.55E+00
WDP*	m³ depriv.	9.85E-01	4.66E-03	9.03E-03	9.99E-01	6.38E-03	2.68E-03	1.94E-01	1.85E+00	9.30E-04	3.81E-03	3.16E-02	3.56E-03	3.09E+00	-8.69E-01
Additional impact															
GWP _{GHG} **	kg CO ₂ eq.	5.13E+00	8.42E-02	6.93E-01	5.91E+00	1.02E-01	4.24E-02	6.46E-01	1.60E+01	8.05E-03	6.18E-02	7.21E-02	1.92E-02	2.29E+01	-8.10E-01

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks. The method Environmental Footprint 3.1 was applied for the assessment.

GWP _{tot}	Climate change total	ADPE	Depletion of abiotic resources – minerals and metals
GWP _{fos}	Climate change – fossil	ADPF	Depletion of abiotic resources – fossil fuels
GWP _{bio}	Climate change – biogenic	WDP	Water use
GWP _{luluc}	Climate change – land use and land use change	GWP _{GHG}	Climate change - greenhouse gas
ODP	Ozone Depletion	* The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator. **The indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. Thus, this indicator is almost equal to the GWP indicator originally defined in EN 15804:2012+A1:2013.	
AP	Acidification		
EP _{fw}	Eutrophication aquatic freshwater		
EP _{mar}	Eutrophication aquatic marine		
EP _{ter}	Eutrophication terrestrial		
POCP	Photochemical ozone formation		

Impact on natural resources

Use of resources

Material resources are based on specific data of the product, i.e. new and replacement material, packaging, and auxiliary materials used in the manufacturing.

Energy resources are calculated based on measurements or LCI-data. All data has been extended to their life cycle scope.

Table of results – use of resources UC 3 per tkm															
	EN15804	Product stage				Construction process stage		Use stage		End-of-life stage					Net Benefits
Impact category	Unit	A1	A2	A3	Sum A1–A3	A4	A5	B2	B6	C1	C2	C3	C4	Total	D
PERE	MJ	3.63E+00	1.41E-02	3.89E+00	7.53E+00	1.84E-02	1.08E-02	1.10E+00	1.87E+01	9.41E-03	1.40E-02	1.01E-01	1.73E-03	2.75E+01	-8.40E-01
PERM	MJ	1.97E-01	0.00E+00	6.37E+00	6.57E+00	0.00E+00	-4.46E+00	5.71E-02	0.00E+00	0.00E+00	0.00E+00	-8.95E-02	0.00E+00	2.08E+00	0.00E+00
PERT	MJ	3.83E+00	1.41E-02	1.03E+01	1.41E+01	1.84E-02	-4.45E+00	1.16E+00	1.87E+01	9.41E-03	1.40E-02	1.19E-02	1.73E-03	2.96E+01	-8.40E-01
PENRE	MJ	5.67E+01	1.17E+00	6.66E+00	6.46E+01	1.45E+00	3.52E-01	8.01E+00	1.56E+02	7.84E-02	8.49E-01	6.14E-01	8.37E-02	2.32E+02	-7.55E+00
PENRM	MJ	1.17E+00	0.00E+00	2.84E-01	1.45E+00	0.00E+00	-1.99E-01	3.55E-01	0.00E+00	0.00E+00	0.00E+00	-4.49E-01	0.00E+00	1.16E+00	0.00E+00
PENRT	MJ	5.79E+01	1.17E+00	6.95E+00	6.60E+01	1.45E+00	1.53E-01	8.36E+00	1.56E+02	7.84E-02	8.49E-01	1.65E-01	8.37E-02	2.33E+02	-7.55E+00
SM*	kg	5.88E-01	0.00E+00	9.12E-04	5.89E-01	0.00E+00	0.00E+00	1.17E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.00E-01	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	7.19E-04	7.19E-04	0.00E+00	0.00E+00	5.75E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.24E-04	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	7.19E-04	7.19E-04	0.00E+00	0.00E+00	5.75E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.24E-04	0.00E+00
FW	m³	2.81E-02	1.03E-04	7.02E-04	2.89E-02	1.43E-04	1.14E-04	5.27E-03	5.21E-02	2.62E-05	8.61E-05	9.56E-04	8.54E-05	8.77E-02	-1.87E-02

PERE	Use of renewable primary energy excluding renewable energy resources used as raw material	PENRM	Use of non-renewable primary energy resources used as raw material
PERM	Use of renewable primary energy resources used as raw material	PENRT	Total use of non-renewable primary energy resources (primary energy and primary energy resources used as raw material)
PERT	Total use of renewable primary energy resources (primary energy and primary energy resources used as raw material)	SM	Use of secondary material
PENRE	Use of non-renewable primary energy excluding non-renewable energy resources used as raw material	RSF	Use of renewable secondary fuels
		NRSF	Use of non-renewable secondary fuels
		FW	Net use of fresh water

*Average recycled content was considered for metal supply; ferrous metal 30% (World Steel Association), aluminum 74%, copper 20% (ecoinvent).

** For modeling the balance of input and output PERM and PENRM the approach B from Annex 3 of the construction PCR (v1.3.4) was used.

The balance considers the energy used as raw material in product and packaging, minus the usable energy that has left the product system.

The result A-C describes the overall loss of usable energy.

Recognizing value at the end of life

Waste – Categories

Information on waste is given in three categories, considering potential risks from deposition of materials. The highest amount of waste is related to categories with low risk „non-hazardous waste“.

Relevant contributions result from raw material extraction and transformation including mining and processing of metals and from manufacturing.

Table of results – waste categories UC 3 per tkm															
	EN15804	Product stage				Construction process stage		Use stage		End-of-life stage					Net Benefits
Impact category	Unit	A1	A2	A3	Sum A1–A3	A4	A5	B2	B6	C1	C2	C3	C4	Total	D
HWD	kg	7.97E-04	7.59E-06	1.08E-05	8.16E-04	9.35E-06	7.97E-07	5.48E-05	1.12E-04	5.61E-08	5.43E-06	8.16E-07	4.19E-07	9.99E-04	-9.15E-05
NHWD	kg	1.31E+00	4.61E-02	7.84E-02	1.44E+00	7.03E-02	6.61E-03	1.29E-01	1.34E+00	6.75E-04	2.68E-02	1.02E-02	4.65E-01	3.49E+00	-2.42E-01
RWD	kg	5.13E-05	2.20E-07	8.44E-06	6.00E-05	2.92E-07	1.03E-07	1.24E-05	1.68E-04	8.44E-08	2.03E-07	2.25E-07	3.22E-08	2.41E-04	1.14E-06

HWD Hazardous waste disposal RWD Radioactive waste disposal

NHWD Non-hazardous waste disposal

Waste – Output flow

The elevator consists of a high number of materials with recycling potential. Plastic and organic material delivered to municipal incineration were considered for energy recovery. No parts are considered for re-use.

Table of results – environmental output flow UC3 per tkm															
	EN15804	Product stage				Construction process stage		Use stage		End-of-life stage					Net Benefits
Impact category	Unit	A1	A2	A3	Sum A1–A3	A4	A5	B2	B6	C1	C2	C3	C4	Total	D
CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	1.13E-01	1.13E-01	0.00E+00	1.50E-02	4.87E-02	0.00E+00	0.00E+00	0.00E+00	1.88E+00	0.00E+00	2.06E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.32E+00	1.27E-01	0.00E+00	0.00E+00	0.00E+00	1.28E-01	0.00E+00	2.58E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	4.34E+00	2.37E-01	0.00E+00	0.00E+00	0.00E+00	2.39E-01	0.00E+00	4.81E+00	0.00E+00

CRU Components for re-use EE Exported Energy

MFR Materials for recycling EET Exported Energy Thermal

MER Materials for energy recovery

Electricity and district heat in manufacturing (A3) and operation (B6) stage

Electricity and district heat are used during the manufacturing stage from suppliers in different countries. Each country has its own electricity and district heat mix with its own composition and environmental impact. The following table shows the GWP_{GHG} emission factors in kg CO₂ eq./kWh of the country specific supply mix. Chinese supply mix was applied for the operational energy use stage (B6).

Country	Electricity kg CO ₂ eq./kWh	Specification data used	District Heat kg CO ₂ eq./kWh
China	0.97	China supply mix	0.13
Switzerland/FL	0.03	Switzerland residual mix	0.06

Transport to installation site (A4)

Transport from Schindler hub to the installation site in Zhejiang has been used as installation location. A load factor based on ecoinvent 3.9 including empty returns has been considered.

Installation (A5)

For the installation phase, energy, ancillary materials, and packaging waste have been considered.

Means of transport	Distance	Load factor
Truck 16-32 metric tons, EURO 4 (RoW)	198 km	5.79 t

Scenario	Amount
Ancillary materials: Cleaning and degreasing agent	2 l with 0.2 g VOC/l
Water use	0 m³
Energy consumption	21 kWh
Waste material	870 kg

Maintenance (B2)

Maintenance and replacement of worn out parts are necessary to support proper operation over the reference service life. For the commuting of the maintenance personnel, an annual average per installation was applied based on the fleet mileage of the region.

Scenario	Amount	
Preventive maintenance interval	As per component individual plan	
Commuting to installation	0.3 km / year	Passenger car
Commuting to installation	60 km / year	Trolleybus

Preventive maintenance replacement materials	Weight (kg)	Weight (%)
Ferrous metal	93.32	52.15
Non-ferrous metals	6.12	3.42
Plastics and rubbers	47.19	26.38
Inorganic materials	0.70	0.39
Organic materials	0.00	0.00
Lubricants	0.04	0.02
Electric and electronic equipment	5.96	3.33
Batteries and accumulators	25.60	14.31
Total	178.93	100

Energy consumption in operation phase (B6) and energy efficiency classification

Increasing energy efficiency is essential in order to reduce the environmental impact of the elevator and the building. The longest phase in the life cycle is the usage stage, which can extend beyond the 25 years used for EPD calculations, depending on maintenance, replacement of parts and modernization.

Schindler energy efficiency calculation and classification is performed according to ISO 25745-2. The typical usage expectation for a Schindler 5000 is between 200 to 500 trips per day. The classification and estimated annual energy consumption always refer to a specific configuration. Usage, load capacity, energy saving options and site conditions also influence the final rating.

End of life (C2 – C4)

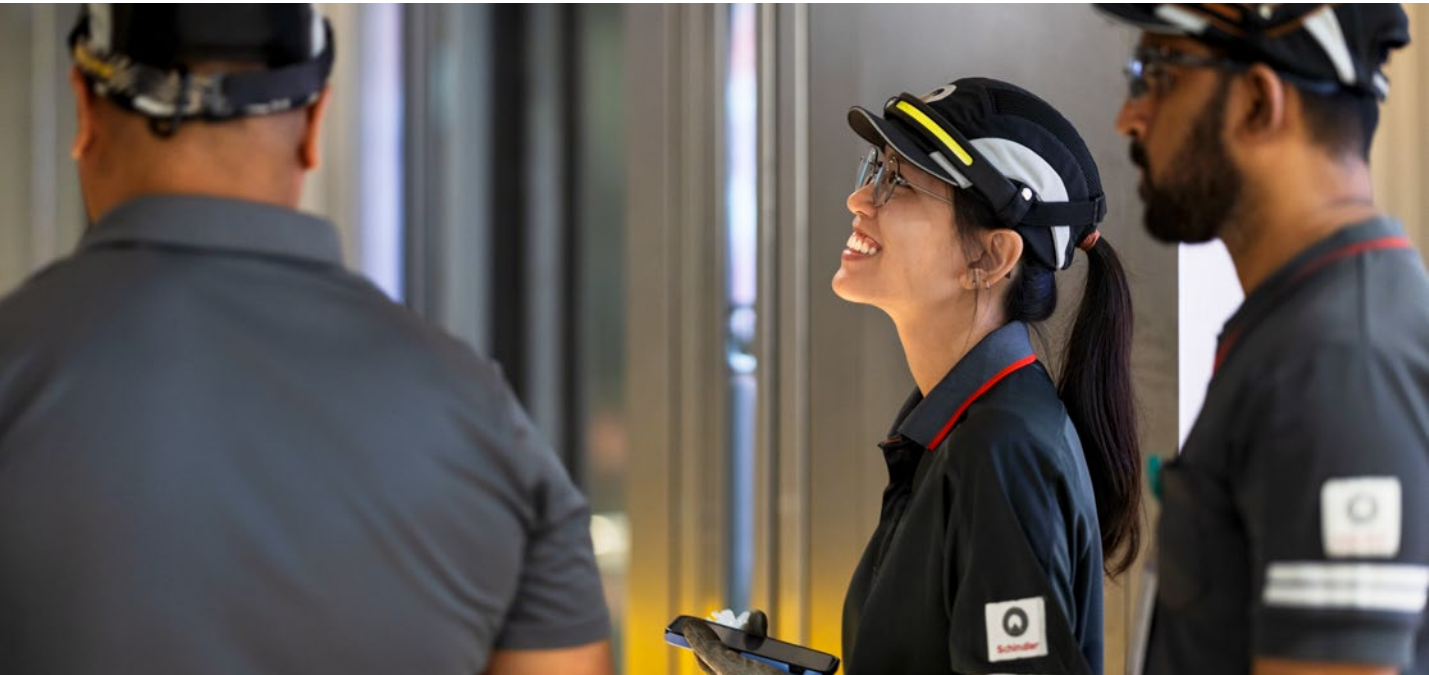
Most materials are suitable for recycling, for example metal and glass, where a recycling rate of 79% is assumed. Plastic and wood are assumed to be disposed of using waste incineration. Energy recovery is assumed standard for municipal waste incineration facilities.

The amount of material delivered to recovery systems is used for the calculations of net benefits in module D. A net flow calculation is used according to EN 15804. Input and outflows of recycled materials are considered.

Usage category	Estimated annual energy consumption	Energy efficiency classification
UC3	1671.4 kWh	Class A

The classification refers to a specific customer configuration as defined for the Life Cycle Assessment, see page 7. Usage pattern, load capacity, customer specific options and site conditions influence the final rating.

Processes	Unit*	Amount kg/kg
Collection process	kg collected separately	1
	kg collected with mixed construction waste	0
Recovery system	kg for re-use	0.00
	kg for recycling	0.79
	kg for energy recovery	0.01
Disposal	kg product or material for final deposition	0.20
Distance for end-of-life treatment	km	45





References

References

ISO 14025:2006 Environmental labels and declarations – Type III environmental declarations. Principles and procedures.

ISO 14040:2006 Environmental management. Life cycle assessment. Principles and frameworks.

ISO 14044:2006 Environmental management. Life cycle assessment. Requirements and guidelines.

GPI v4.0 / v5.0.0 General Programme Instruction for the International EPD® System

EN 15804:2012+A2:2019/AC:2021 Sustainability of construction works – Environmental product declarations – Core rules for the product category of construction products

PCR 2019:14 V1.3.4 Construction Products

C-PCR-008 (TO PCR 2019 :14) V1.0.0 Lifts (Elevators)

ISO 25745-2:2015 Energy performance of lifts, escalators and moving walks - Part 2: Energy calculation and classification for lifts (elevators)

ecoinvent database v3.9, SimaPro V9

Glossary

- LCA – Life Cycle Assessment: Assessment methodology of the environmental impact of all relevant material and energy flows throughout the entire life cycle of a product, according to ISO 14040.
- LCI – Life Cycle Inventory: Creation of inventory of input and output flows for a product system. These flows include inputs such as water, energy, and raw materials. Outputs are releases to air, land, and water. Inventories are based on literature analysis or process simulation.
- EPD – Environmental Product Declaration: A declaration that provides quantified environmental data using predetermined parameters defined in a Product Category Rule, according to ISO 14025.
- PCR – Product Category Rule: A set of specific rules, requirements, and guidelines for developing environmental declarations for one or more product categories.
- REACH – Registration, Evaluation, Authorization and Restriction of Chemicals: EU regulation (EC 1907/2006) that addresses the production and use of chemical substances, and their potential impacts on both human health and the environment.
- RSL – Reference Service Life: The reference service life considered for the LCA corresponds to the definition of C-PCR-008 for a generic lift.
- FU – Functional Unit: For lifts it is defined as the transportation of a load over a distance, expressed as one tonne [t] transported over one kilometer [km], i.e. tonne-kilometer [tkm] over a vertical (or inclined) trajectory.
- UC – Usage Category: Defines the intensity of the lift usage by categories, based on average number of trips per day, according to ISO 25745-2.



Sustainability

We Elevate... Our World

Sustainability at Schindler is more than striving to minimize the use of natural resources. We facilitate sustainable, smart urban mobility, while committing to a sustainable supply chain for all our products and driving innovation for green building management.

Sustainability at Schindler also means enabling an inclusive work environment where our workforce, which is as diverse as our customers and passengers, can thrive. It also means creating value in the communities where we operate by helping develop young talent through education and training, by fostering lifelong learning for our technicians, and by designing products and systems that make it easy and safe for people to move about in cities.

If you would like to learn more about Schindler's sustainability initiatives or access the latest Sustainability Report, you can find them here:
<https://www.schindler.com/com/internet/en/about-schindler/sustainability.html>

This publication is for general informational purposes only and we reserve the right at any time to alter the services, product design and specifications. No statement contained in this publication shall be construed as a warranty or condition, expressed or implied, as to any service or product, its specifications, its fitness for any particular purpose, merchantability, quality or shall be interpreted as a term or condition of any service or purchase agreement for the products or services contained in this publication. Minor differences between printed and actual colors may exist.

We Elevate



Schindler